42.510 Statistical Learning for Data Science

Statistics review

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Week 1, Lecture 2

Probability vs. Statistics

- *Probability* lets us work with models that we can use to generate random variables, quantities and simulated data. For example, we can obtain i.i.d samples from our model. Probability creates a framework for us to model randomness.
- *Statistics* is based on the real world. We work with data sets that we do not know much of, we come up with a set of assumptions about the data set which allows us to return to the probability models so that we can say something concrete about the data set.
- *Learning*: using data to understand real world phenomenon, may or may not always be based on probabilistic assumptions.

Statistics

How we work with data sets in statistics?

1. Assume they are i.i.d observations.
2. Assume they come from a probability distribution F which belongs to a collection of all distributions.
3. We want to infer about F.

- Note that this is *learning* as well, but in statistics we call it *inference*.

Statistics: notations and terminology

- *Random variables*: X, Y, Z.
- *Realizations, observations, observed values*: x, y, z.
- *Data*: A collection of realization of random variables.
- *Sample*: Random sample of size n: $X_1, \ldots, X_n$ iid from some distribution F.
- *Realization of a random sample*: $x_1, \ldots, x_n$.
- *Statistic*: a function of the data.
- *Parameter*: true value that numerically characterizes a population value. This is often not possible to know exactly in practice. A statistic is often used to estimate such parameters.
- *Sample mean*: $\overline{x}_n$ where $x_1, \ldots, x_n$ are realizations of $X_1, \ldots, X_n$.
- We may call $\overline{X}_n$ to be sample mean as well.

Descriptive statistics formulas

- Let $(X, Y) \sim F$ with $X \sim F_1, Y \sim F_2$.
- Let $(X_1, Y_1), \ldots, (X_n, Y_n)$ be iid from F.
- Let $(x_1, y_1), \ldots, (x_n, y_n)$ be realizations from the above sample.

Name	Parameter	Statistic (realization)
Mean	$\mu_X = \mathbb{E}(X)$	$\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$
Variance	$\sigma_X^2 = \mathrm{Var}(X)$	$s_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2$
Std. dev.	$\sigma_X = \sqrt{\mathrm{Var}(X)}$	$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2}$
Covariance	$\sigma_{XY} = \mathbb{E}(X - \mathbb{E}(X))(Y - \mathbb{E}(Y))$	$s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)(y_i - \bar{y})$
Correlation	$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$	$r_{xy} = \frac{s_{xy}}{s_x s_y}$

Estimating the number of German tanks

- During World War II, the Allied forces had noted the serial numbers of some captured/destroyed German tanks and wanted to estimate the total number of German tanks deployed.
- They correctly guessed that the serial numbers ran from 1 to N, where N was the total number of tanks produced.
- **QUESTION:** How can one estimate N?

Month of production	Intelligence estimate	Statistical estimate	Actual (from post-war captured documents)
June 1940	1000	169	122
June 1941	1550	244	271
August 1942	1550	327	342

Estimating the number of German tanks

- **Goal:** Estimate the value of N from a sample of discrete uniform random variable taking values in $\{0, 1, \ldots, N\}$. We are given a sample of size n drawn without replacement.
- Let the sample observations be $\{x_1, \ldots, x_n\}$.
- Order the sample: $x_{(1)} \leq \ldots \leq x_{(n)}$.
- Some possible estimates for N are:
  1. $\bar{x}_n + 3s_x$
  2. $2\bar{x}_n - 1$
  3. $x_{(n)}$
  4. $x_{(n)} + x_{(1)} - 1$
  5. $\frac{n+1}{n} x_{(n)} - 1$

- Estimate N if the numbers you got are:
  - 588; 601; 126; 699; 464; 468; 258; 440; 275; 320

- **R exercise:** Draw 500 simulated samples, each of size $n = 5$, from a population with $N = 250$. Evaluate the estimators. Compare variability, bias.

Simulation: R code

```r
T<- 250 ## actual number of tanks
k<- 5 ## number of tanks seen
n<- 500 ## number of simulations

## n simulations of sample size k from 1,...,T
tanks<-matrix(replicate(n,sample(1:T,k)),nrow=n,ncol=k,byrow = "TRUE")

## Initialize matrices for storing the estimates
Ests<-matrix(rep(0,n*k),nrow=n,ncol=k)

Ests[,1]<- apply(tanks,1,function(x) mean(x)+3*sqrt(var(x))) ## Estimator 1
Ests[,2]<- apply(tanks,1,function(x) 2*mean(x)-1)            ## Estimator 2
Ests[,3]<- apply(tanks,1,function(x) max(x))                 ## Estimator 3
Ests[,4]<- apply(tanks,1,function(x) max(x)+min(x)-1)        ## Estimator 4
Ests[,5]<- apply(tanks,1,function(x) (k+1)*max(x)/k-1)       ## Estimator 5

mean_ests<-apply(Ests,2, function(x) mean(x))                ## Mean of the estimates

sd_ests<-apply(Ests,2,function(x) sd(x))                     ## Sd of the estimates

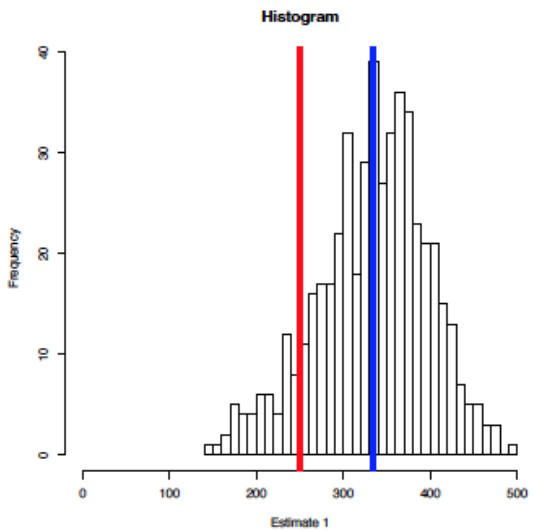

par(mfrow=c(2,3))
for(i in 1:5)
{hist(Ests[,i],density=FALSE,freq=T,breaks = 30, main="Histogram",xlab=paste("Estimate",i),xlim=c(0,500))
abline(v=250,col="red",lwd=3)
abline(v=mean_ests[i],col="blue",lwd=3)
}

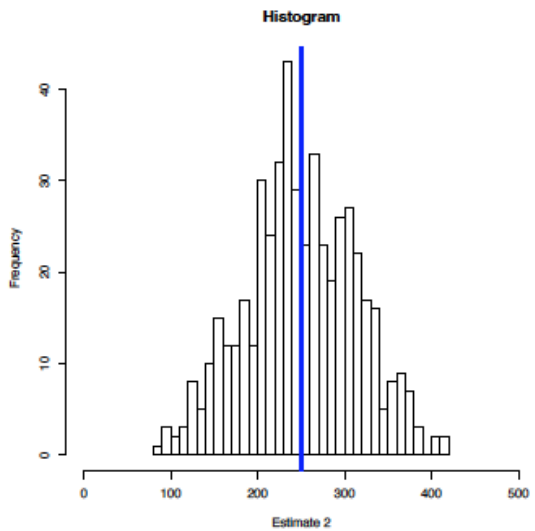

dft<-data.frame(est=c("Est1","Est2","Est3","Est4","Est5"), mean=mean_ests,sd=sd_ests)

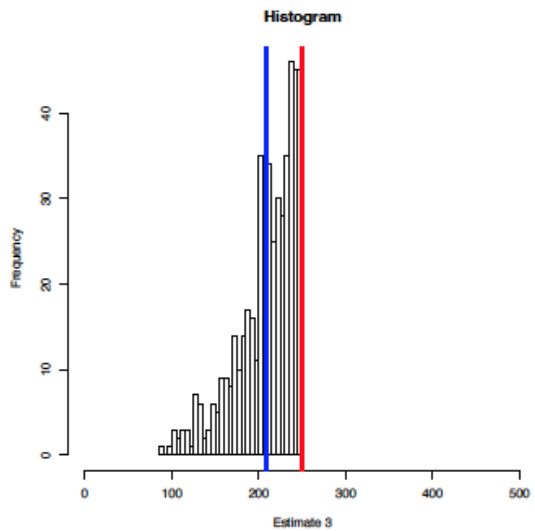


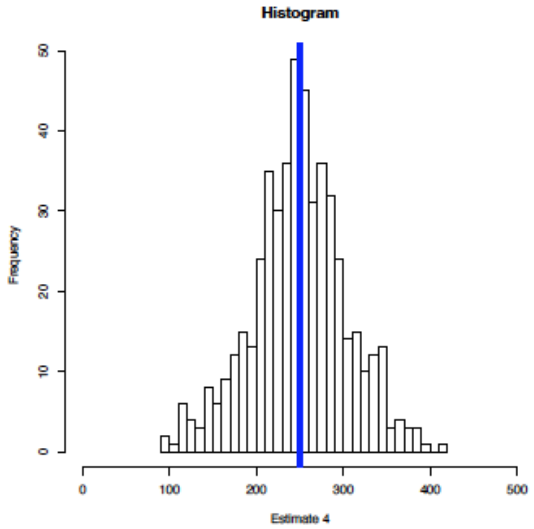
print(dft,digits=5)
```

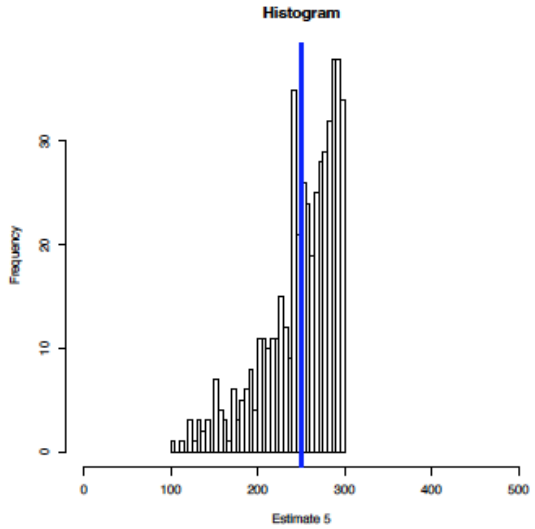
Estimating the number of German tanks











- Histogram from a run

Estimating the number of German tanks

	est	mean	sd
1	Est1	334.21	64.06
2	Est2	250.06	63.54
3	Est3	209.10	34.34
4	Est4	249.82	54.78
5	Est5	249.92	41.21

- Estimates from the run.
- **Exercise:** Theoretically analyze the five different estimators with respect to variance, bias and compare.

Estimating the number of German tanks

- Estimate N if the numbers you got are:
  - $588; 601; 126; 699; 464; 468; 258; 440; 275; 320$
- Some possible estimates for N are:
  1. $\bar{x} + 3s$ $\quad (\hat{N}_1 = 960)$
  2. $2\bar{x} - 1$ $\quad (\hat{N}_2 = 847)$
  3. $x_{(n)}$ $\quad (\hat{N}_3 = 699)$
  4. $x_{(n)} + x_{(1)} - 1$ $\quad (\hat{N}_4 = 824)$
  5. $\frac{n+1}{n} x_{(n)} - 1$ $\quad (\hat{N}_5 = 768)$
- I chose $N = 735$.

Key concepts in Statistical inference

- Given observations/ realizations $x_1, \ldots, x_n$: data, assume that they are i.i.d. realizations from $X_1, \ldots, X_n$ from some distribution F.

Questions

- Can we know F? What can we say about F?
  - Parametric approach: assume Normal, exponential, binomial and estimate the parameters.
    - Here again we may have frequentist and Bayesian approaches.
  - Non-parametric approach: no specific parametric model, estimate from the entire data
- Can we infer about specific functionals of F: mean, median, variance, skewness, etc.

What do we do in statistical inference?

- Point estimation
- Interval estimation
- Hypothesis testing

Point estimation

- Point estimate: single best estimate for a parameter, c.d.f, regression function, etc.
- Estimate vs Estimator
  - Estimate
    - ★ Point estimate of θ is denoted by $\hat{\theta}_n$ or $\hat{\theta}$.
    - ★ $\hat{\theta}$ is obtained from the observations (varies when you change the dataset).
    - ★ Of course, θ is a fixed unknown parameter.
  - Estimator
    - ★ Point estimator is a function of the random sample: $X_1, \ldots, X_n$.
    - ★ $\hat{\theta}_n = g(X_1, \ldots, X_n)$. This is a random variable.
- Methods of estimation (parametric)
  - Maximum likelihood estimation (MLE).
  - Method of moments: *match theoretical moments with empirical moments*.

Bias and variance trade-off

- Let $\hat{\theta}_n$ be an estimator for θ. We want to assess the quality of the estimator. (keep in mind our example)
  1. Bias$(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n) - \theta$.
     - An estimator $\hat{\theta}_n$ is unbiased if Bias$(\hat{\theta}_n) = 0$.
  2. Standard error of $\hat{\theta}_n$ (variability of the estimator):
     - $\text{se}(\hat{\theta}_n) = \sqrt{\text{Var}(\hat{\theta}_n)} = \sqrt{\mathbb{E}(\hat{\theta}_n - \mathbb{E}(\hat{\theta}_n))^2}$
  3. Mean Squared Error (MSE)
     - $\text{MSE}(\hat{\theta}_n) = \mathbb{E}(\hat{\theta}_n - \theta)^2$.

Theorem

$$MSE(\hat{\theta}_n) = (Bias(\hat{\theta}_n))^2 + Var(\hat{\theta}_n).$$

Consistency of point estimators

- A point estimator $\hat{\theta}_n$ is consistent for θ if as $n \to \infty$, either
  - $\hat{\theta}_n \xrightarrow{P} \theta$ (weak consistency)
  - $\hat{\theta}_n \xrightarrow{a.s.} \theta$ (strong consistency)

Theorem

If $Bias(\hat{\theta}_n) \to 0$ *and* $se(\hat{\theta}_n) \to 0$ *as* $n \to \infty$ *then* $\hat{\theta}_n \xrightarrow{P} \theta$.

- We can check this for the previous example (and later ones).

Maximum Likelihood Estimation (MLE)

- Introduced by R.A. Fisher in the 1920s.
- The idea is to pick the value of the parameter which maximizes the *probability* of observing our data.
- For example, suppose we want to estimate the parameter p of a Bernoulli distribution. A random sample of size 5 drawn from this distribution reads:

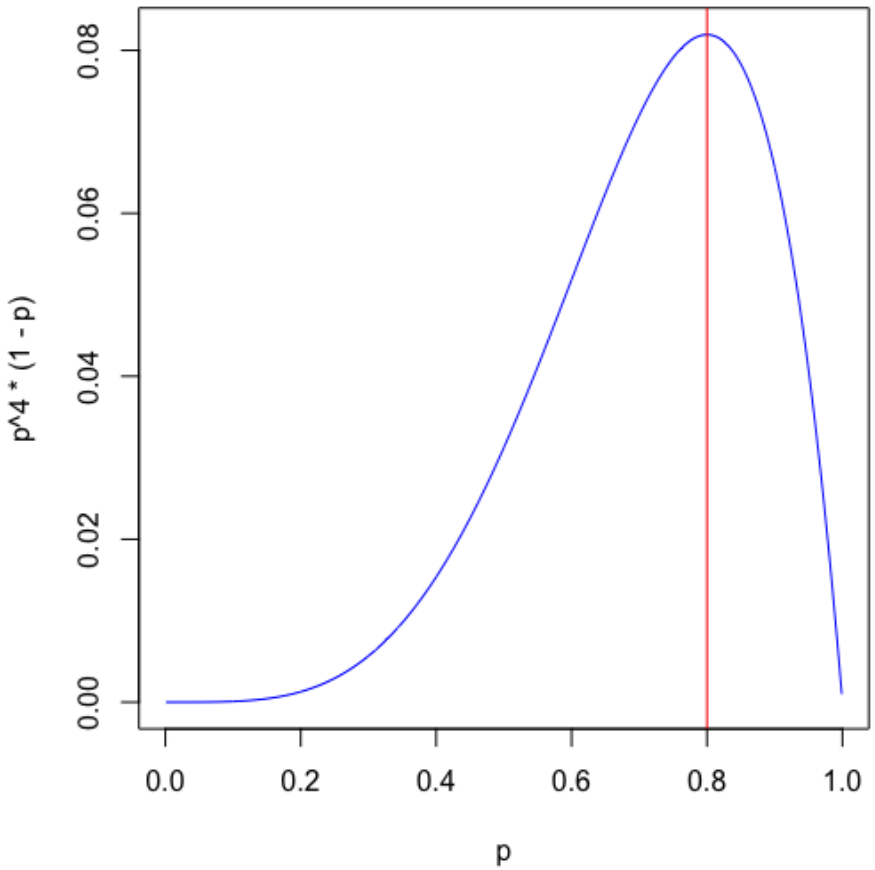
$$1, 1, 1, 0, 1.$$

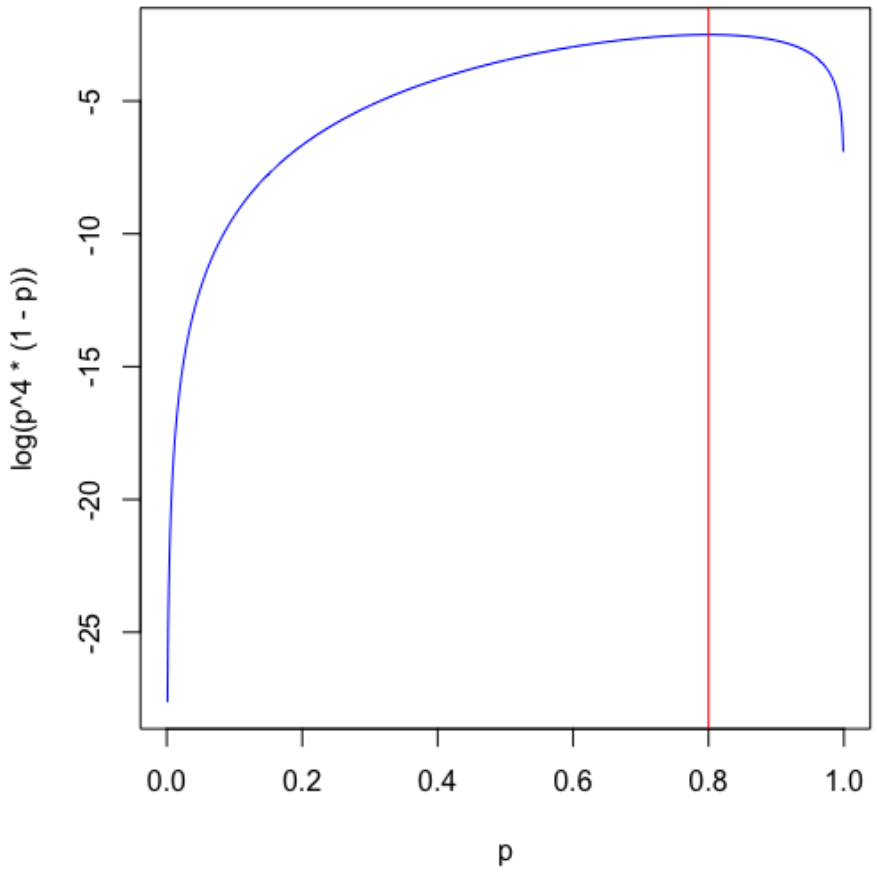
- Which of these is most likely?

$$p = 0; p = 1/2; p = 2/3; p = 4/5; p = 1.$$

- Probability of this event $= p^4(1 - p)$.

MLE: Likelihood and log-likelihood





Likelihood function

- Given a probability distribution, let θ be a parameter to be estimated, and denote the probability density (or mass) function by $f(x|\theta)$.
- The **likelihood function** of θ, called $L(\theta)$ is the joint pdf (density) for a random sample of size n with observations $x_1, x_2, \ldots, x_n$, is

$$L(\theta) := f(x_1, \ldots, x_n|\theta) = \prod_{i=1}^{n} f(x_i|\theta).$$

- If the maximum of this function occurs at $\widehat{\theta}$, then $\widehat{\theta}$ is the value of the parameter which maximizes the probability of observing our data, and we will use it as our estimate for θ.

Maximum likelihood estimate

- In many cases, we can find $\widehat{\theta}$ by solving for

$$\frac{\mathrm{d}}{\mathrm{d}\theta} L(\theta) = 0, \quad \text{or equivalently,} \quad \frac{\mathrm{d}}{\mathrm{d}\theta} \ln(L(\theta)) = 0.$$

- We can use the *2nd derivative* to check that it is a maximum.
- Example: let $x_1, \ldots, x_n$ be a random sample from Bernoulli(p). Then

$$L(\theta) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{n\bar{x}}(1-p)^{n-n\bar{x}}.$$

  - Solving for $\frac{\mathrm{d}}{\mathrm{d}\theta} \ln(L(\theta)) = 0$, we find that $\widehat{p} = \overline{x}$.

- **Exercise:** Find the MLEs for μ and σ^2 where $x_1, \ldots, x_n$ are observations from a normal distribution with parameters μ and σ^2. The normal pdf is:

$$f(x|\mu, \sigma^2) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty.$$

  Are the estimators unbiased?

Adult persistence in head-turning asymmetry

- A neonatal right-side preference makes surprising romantic reappearance later in life.
- p = proportion of couples turning head to the right while kissing.
- Device an experiment to test if there is any truth in the statement.

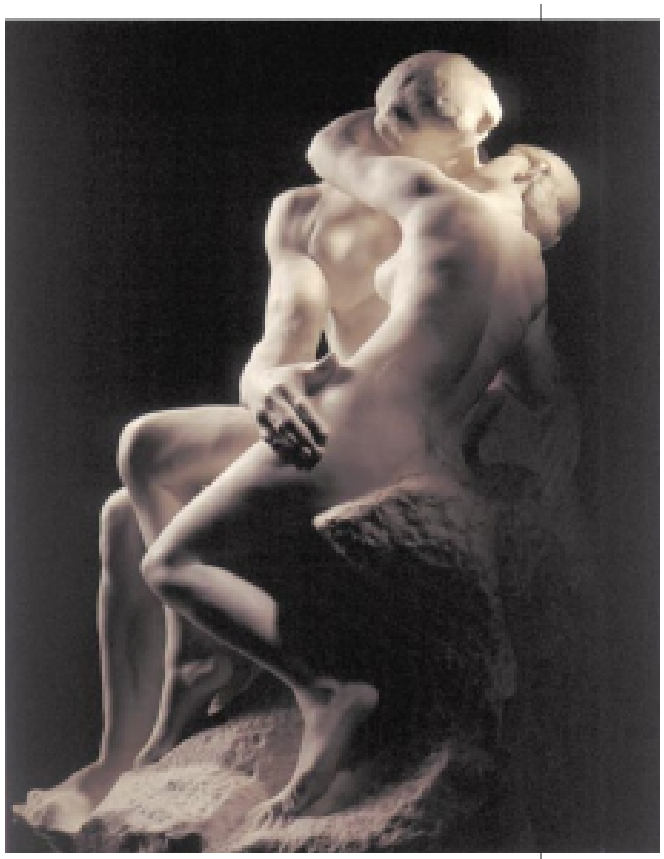


Figure 2 *The Kiss*: the couple in Auguste Rodin's masterpiece are turning their heads to the right to kiss each other.

Human behaviour: Adult persistence of head-turning asymmetry, Nature, 2003. Nature article

Adult persistence in head-turning asymmetry

- Observe $n = 124$ couples while kissing.
- Collect data: $1 =$ right, $0 =$ left.
- Observed in study:
  - 80 out of 124 couples kissed from right.
- Estimate

$$\widehat{p} = \frac{80}{124} = 64.5\%.$$

- Is there a preference to the right?

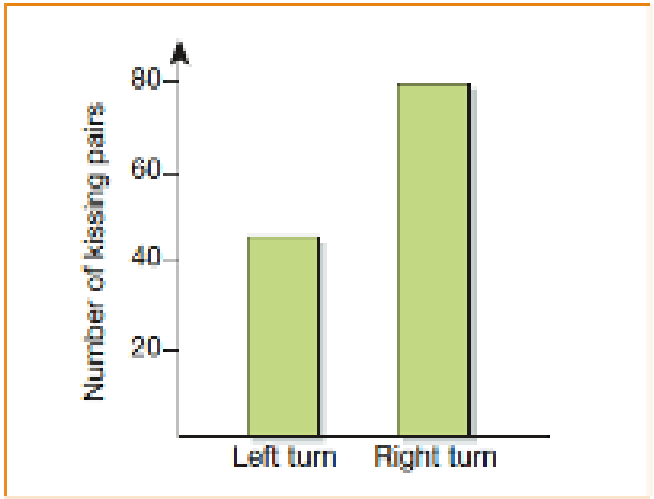


Figure 1 The number of couples who turn their heads to the right rather than to the left when kissing predominates by almost 2:1 (64.5%: 35.5%; $n = 124$ couples).

- Suppose $n = 3$ and you observe *right*, *right*, *left*. Is there a preference to the right?

Adult persistence in head-turning asymmetry

Modeling

- $n = 124$.
- Define $X_i = \begin{cases} 1 & \text{if couple kisses from right,} \\ 0 & \text{otherwise.} \end{cases}$

Assumptions

- X_i's are random variables.
- X_i's are Bernoulli(p).
- X_i's are independent.

Consequence

- **WLLN**: $\widehat{p} = \overline{X}_n \xrightarrow{P} p$, hence p is close to $\hat{p} = 0.645$. (point estimation)
- But how close? Can we be confident that $p \neq 0.5$?

Central limit theorem: confidence intervals (interval estimation)

- Using Central limit theorem (**CLT**) we have as $n \to \infty$:

$$\frac{\sqrt{n}(\overline{X}_n - p)}{\sqrt{p(1-p)}} \xrightarrow{d} \mathcal{N}(0,1).$$

- Hence for large n:

$$\overline{X}_n \approx \mathcal{N}\left(p, p(1-p)/n\right).$$

- If $Z \sim \mathcal{N}(0,1)$ and we define z_α such that $\mathbb{P}(Z \leq z_\alpha) = 1-\alpha$, then

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\overline{X}_n - p)}{\sqrt{p(1-p)}} \leq z_{\alpha/2}\right) \approx 1-\alpha.$$

$$\text{or, } \mathbb{P}\left(\overline{X}_n - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} \leq p \leq \overline{X}_n + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right) = 1-\alpha.$$

Adult persistence in head-turning asymmetry

Central limit theorem: confidence interval (interval estimation)

- We know $p(1-p) \leq \frac{1}{4}$. Hence with $(1-\alpha)100\%$ confidence we can say that:

$$p \in \left[\overline{X}_n - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}, \overline{X}_n + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}\right] \subseteq \left[\overline{X}_n - \frac{z_{\alpha/2}}{2\sqrt{n}}, \overline{X}_n + \frac{z_{\alpha/2}}{2\sqrt{n}}\right]$$

- Consider $\alpha = 0.05$, then $z_{0.025} = 1.96$. Hence with at least 95% confidence, we have approximately $p \in [0.56, 0.73]$.
- What happens when $n = 3$?
- We can't use CLT, but even with CLT, with 95% confidence we can say that $p \in [0, 1]$.
- Check *Hoeffding bounds* and use them to get a more useful result here.

Adult persistence in head-turning asymmetry

Central limit theorem: hypothesis testing

- We can also conduct a **hypothesis test** here.
- Suppose we actually had $p = \frac{1}{2}$. Then **CLT** will tell us

$$\mathbb{P}\left(\frac{\sqrt{n}\left|\overline{X}_n - \frac{1}{2}\right|}{\sqrt{\frac{1}{2} \times \frac{1}{2}}} \leq z_{\alpha/2}\right) \approx 1 - \alpha, \quad \text{or, } \mathbb{P}\left(2\sqrt{n}\left|\overline{X}_n - \frac{1}{2}\right| > z_{\alpha/2}\right) \approx \alpha$$

- So reject the premise that "$p = \frac{1}{2}$" if $\left|\overline{X}_n - \frac{1}{2}\right|$ is a very large number.
- In statistical terminology, at level $\alpha = 0.05$, reject "$p = \frac{1}{2}$" if $\left|\overline{X}_n - \frac{1}{2}\right| > \frac{z_{\alpha/2}}{2\sqrt{n}}$.
- With $\alpha = 0.05$, here we have $z_{0.025} = 1.96, n = 124$ and hence

$$\left|\overline{X}_n - \frac{1}{2}\right| = |0.645 - 0.5| = 0.145 >> 0.09 = \frac{1.96}{2\sqrt{124}}.$$

- Hence we reject the premise or *hypothesis that* "$p = \frac{1}{2}$" at level 0.05.

Hypothesis testing for means: chart

Purpose	Assumption on data	Test used	Formula	Distribution of statistic under H_0	Notations
Testing if sample comes from a population with a fixed hypothesized **mean** value	• Normal with **known** variance • **Or,** the sample size ***n* is large** for CLT to hold (n>>30)	One sample z-test	$z = \frac{(\bar{x} - \mu)}{\frac{\sigma}{\sqrt{n}}}$	Normal	• $\bar{x}$ is sample mean • μ is population average • σ is population s.d. • s is sample s.d. (sometimes for large samples s replace σ for the z-test when σ is unknown) • n is sample size
	• Normal with **unknown** variance (small sample)	One sample t-test	$t = \frac{(\bar{x} - \mu)}{\frac{s}{\sqrt{n}}}$	t_{n-1}	
Testing for equality (or a particular value of difference) of mean between Population 1 and Population 2.	• Both populations are Normal with **known** and not necessarily **equal** variance **AND,** both sample sizes are **large** for CLT to hold (n_1>>30, n_2>>30)	Two sample z-test	$z = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	Normal	• $\bar{x}_1, \bar{x}_2$ are sample means • μ_1, μ_2 are population averages • σ_1, σ_2 are population s.d., may be replaced by s_1, s_2 for large samples when σ_1, σ_2 are unknown • n_1, n_2 are sample sizes
	• Both populations are Normal with **unknown** and not necessarily **equal** variance **AND,** both sample sizes are **small.**	Welch's (unequal variance) t-test	$t = \frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	t_v	The statistic follows an approximate t-distribution with d.f: $v \approx \frac{\left(\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)^2}{\left(\frac{\sigma_1^4}{n_1^2(n_1-1)} + \frac{\sigma_2^4}{n_2^2(n_2-1)}\right)}$
	• Both populations are Normal with **unknown** and **equal** variance **AND,** both sample sizes are **small.**	Two sample t-test	$\frac{(\bar{x}_1 - \bar{x}_2)}{\sqrt{S_p^2 * (\frac{1}{n_1} + \frac{1}{n_2})}}$	$t_{n_1+n_2-2}$	In addition to above • $S_p^2 = \frac{(n_1-1)*S_1^2 + (n_2-1)*S_2^2}{n_1+n_2-2}$ • S_p^2 is their pooled variance or estimator of their common sd
	• The populations are **paired or dependent** (and naturally with same sample size).	Paired t-test	$t = \frac{\bar{d}}{\left(\frac{s}{\sqrt{n}}\right)}$	t_{n-1}	✓ $\bar{d}$ is average of the difference between the elements of the two samples ✓ s is the std. dev. of the difference between the elements of the two samples: $s = \sqrt{\frac{\sum_{i=1}^{n} d_i^2 - \bar{d}^2}{(n-1)}}$

Testing and distribution families

These are for you to explore.

- t-distribution and t-test (vs z-test).
- χ^2-distribution and χ^2-test of independence (non-parametric).
- F-distribution and F-test.

Student's t-distribution

- Defined on the entire real line like Normal distribution.
- Looks like normal (but heavier-tailed).
- Has one parameter which is positive integer-valued, which is called the **degree of freedom.**

Snedecor's F-distribution

- Defined on the non-negative reals.
- Has two parameters: they are positive integer-valued and also called degrees of freedom (d.f.).

References

- *Class notes* for Statistical learning for Data Science (2025).
- Wasserman (2004) *All of Statistics: a concise course in statistical inference (Vol. 26)*.